

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis AKD Kollmorgen Module R&R	Effective Date	1 May 2019

This procedure is applicable to below models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL		√	√
V810 / V810i S2EX		√	√
V810 / V810i Mini			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	1 May 2019	First Release	TAN LIANG KAI

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X-Axis AKD Kollmorgen Part ID : 4XASPP-A106

X-Axis Kollmorgen Removal

1. Unload board/panel from V810 machine if any.
2. Go to Connected Processor tab with V810 GUI, click "Stop RMS".

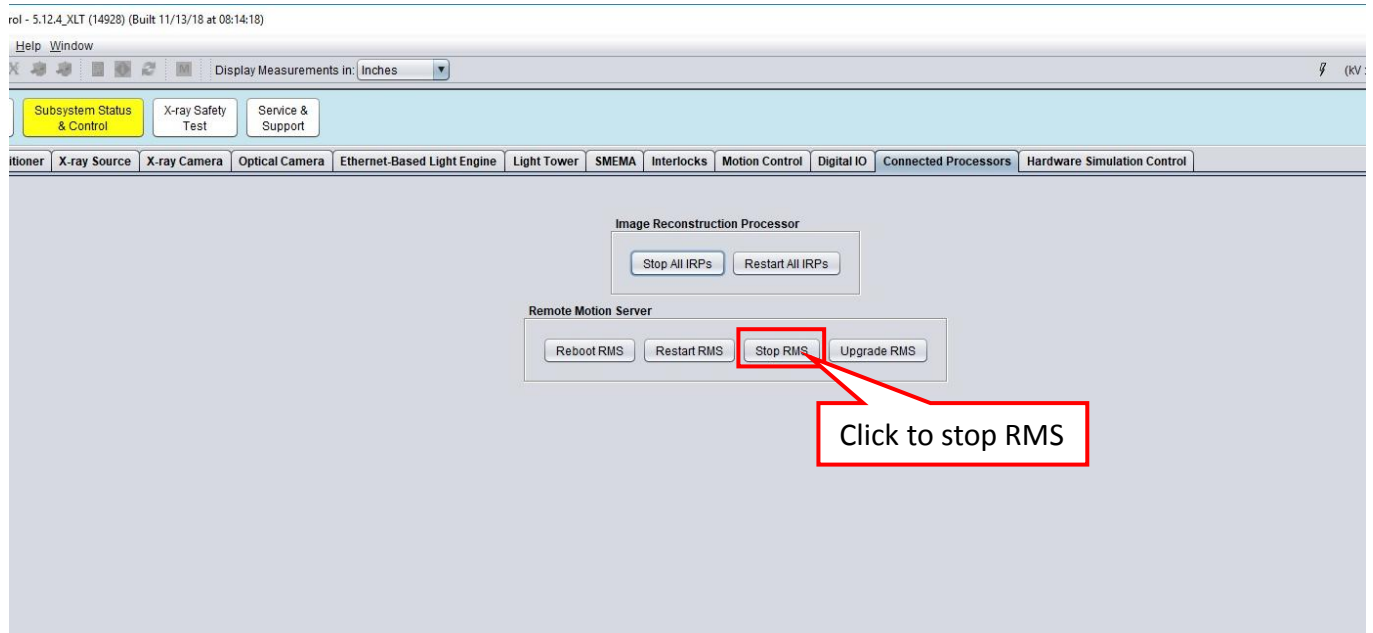


Figure 1 Stop RMS Driver

3. Turn off X-Ray source & SSCA power.
4. Open SSCA top cover.
5. Identify X-Axis AKD Kollmorgen location by refer below Figure.

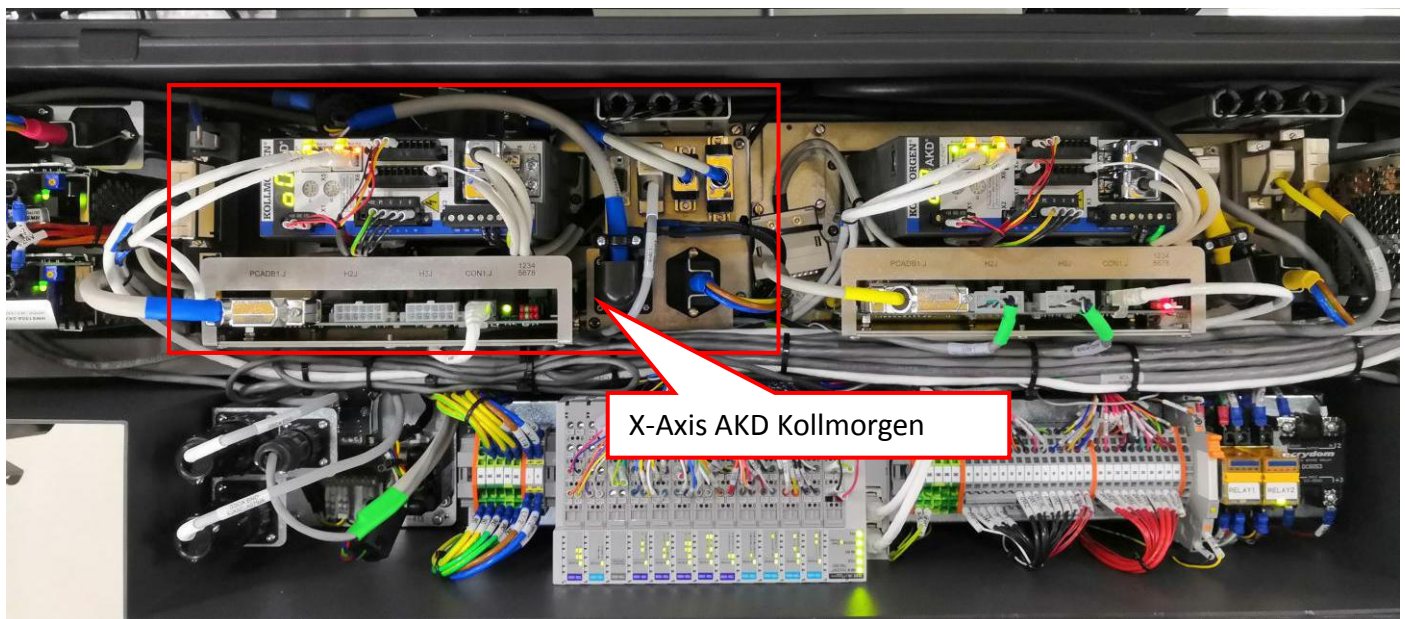


Figure 2 SSCA

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6. Disconnect all cables from X-Axis AKD Kollmorgen.

Remark:

- i. Each cable has its orientation & secure lock. Remember to unlock first before disconnect the cable.
- ii. Perform marking on each cable before disconnect.

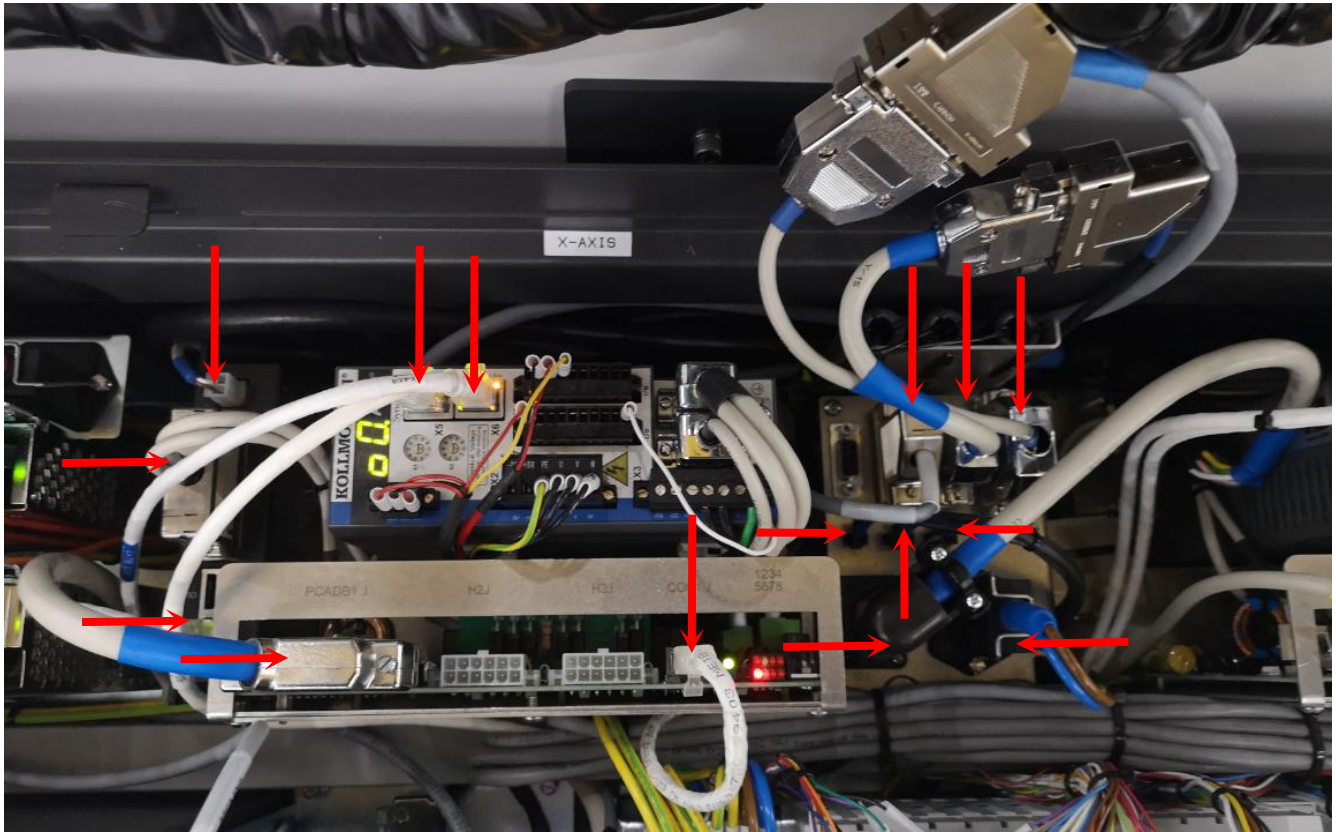


Figure 3 X-Axis AKD Kollmorgen

7. Loosen 4x screws at X-Axis AKD Kollmorgen module support base.

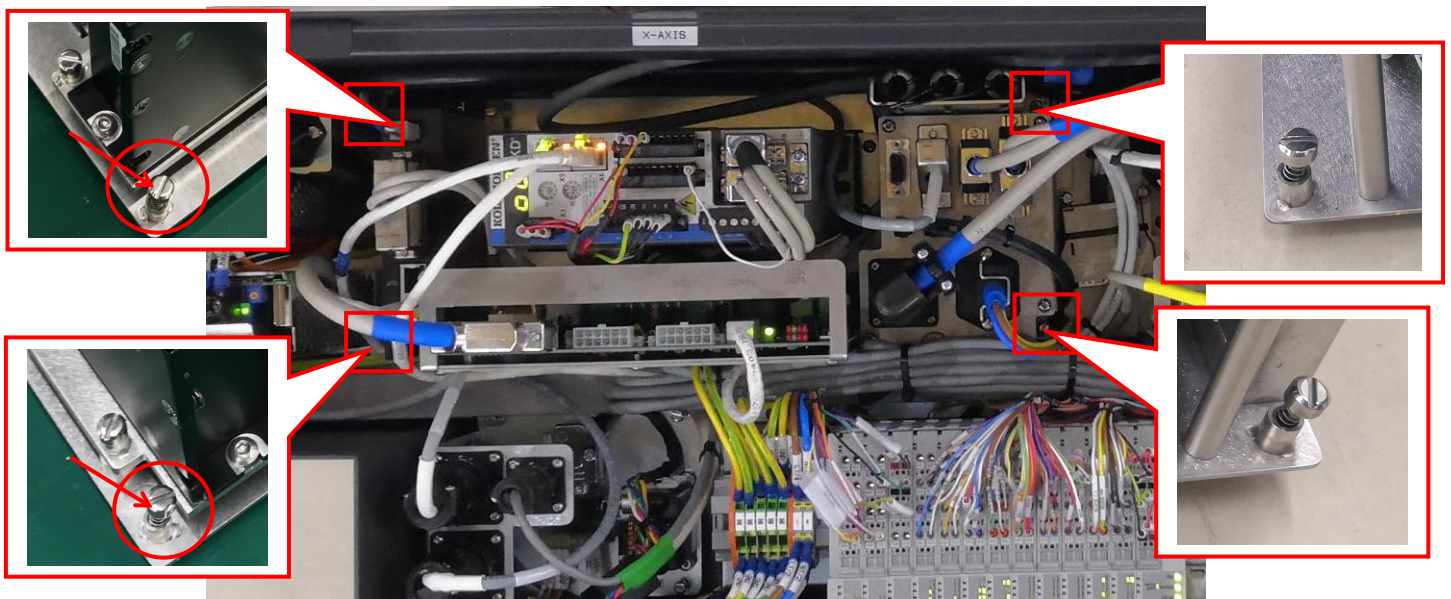


Figure 4 X-Axis AKD Kollmorgen

8. Remove X-Axis AKD Kollmorgen module from SSCA.

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STC Driver Removal

1. From old X-Axis AKD module, disconnect cable from STC driver.
2. Unscrew and remove STC Driver.

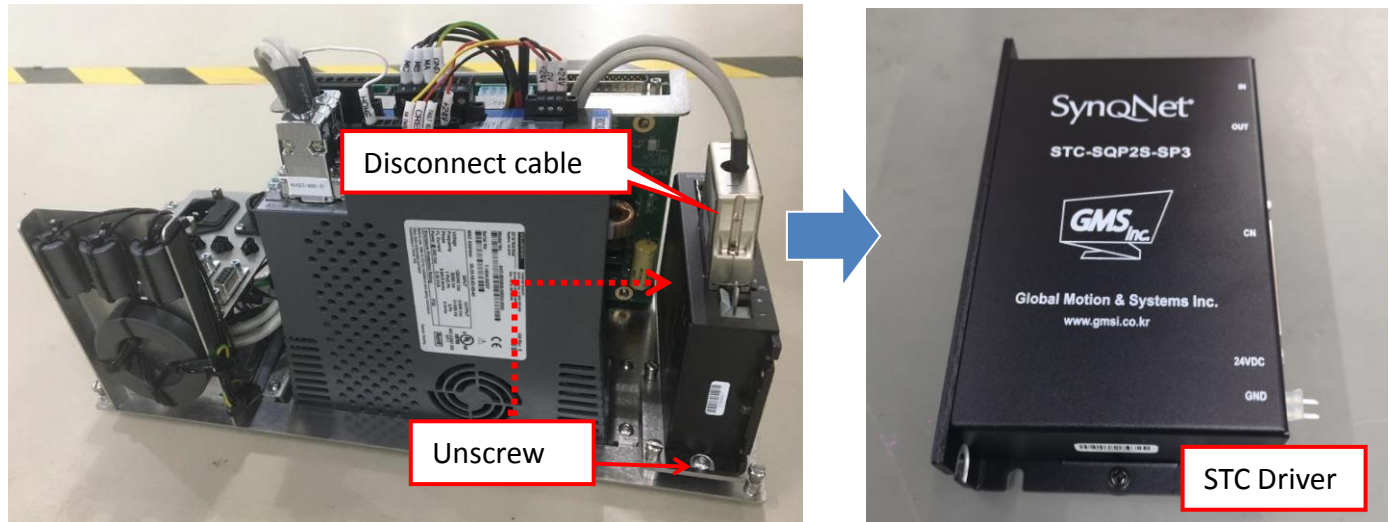


Figure 5 Remove STC Driver

STC Driver Replacement

1. Assemble STC Driver to new X-Axis AKD Kollmorgen.
2. Connect cable back to STC Driver & tighten up 2x screws.

Remark: Using cable tie to tidy up cables.

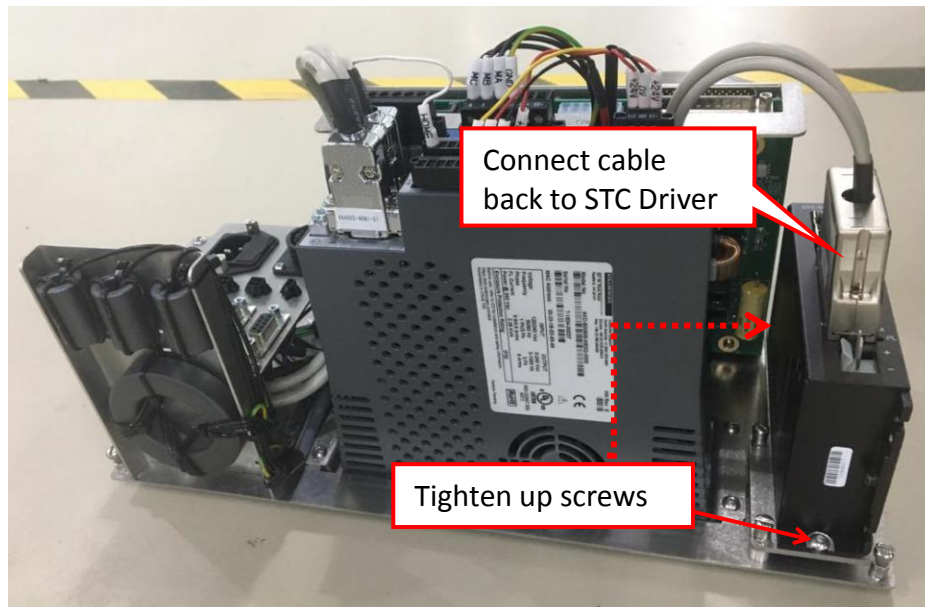


Figure 6 Assemble STC Driver

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X-Axis Kollmorgen Replacement

1. Assemble X-Axis AKD Kollmorgen to SSCA.
2. Tighten 4x screws at X-Axis AKD Kollmorgen module support base.

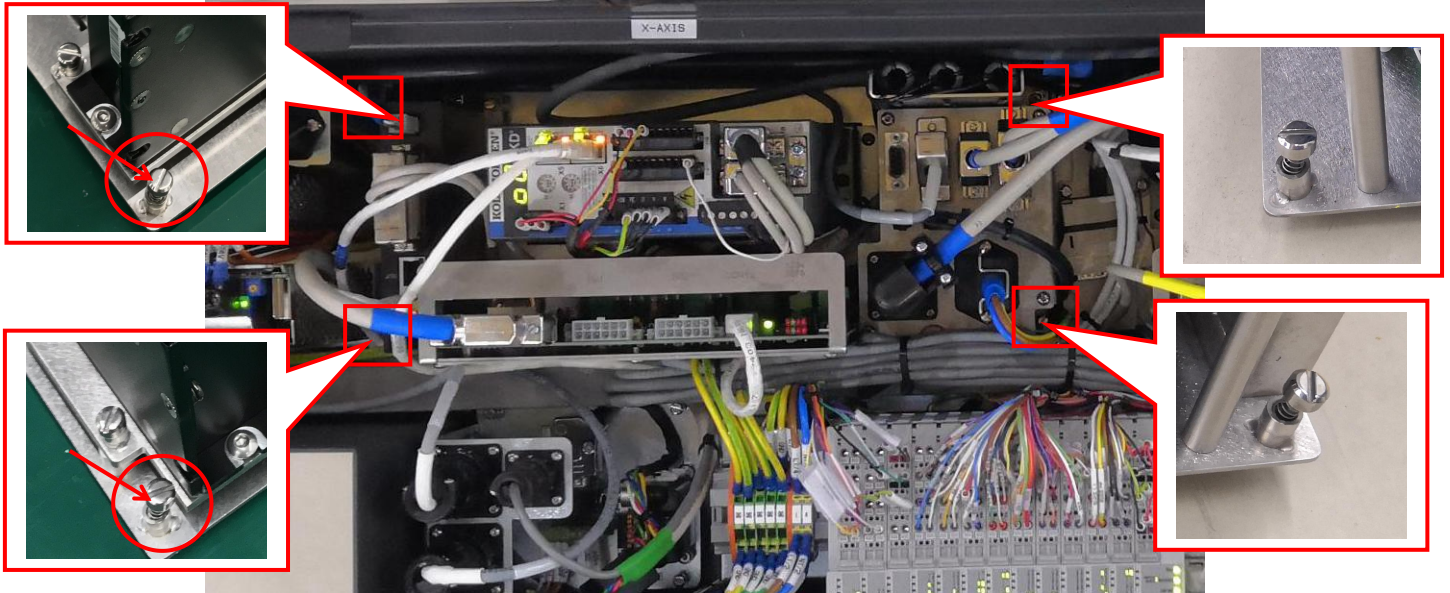


Figure 7 X-Axis AKD Kollmorgen

3. Refer to cable marking & connect all cables back to X-Axis AKD Kollmorgen.

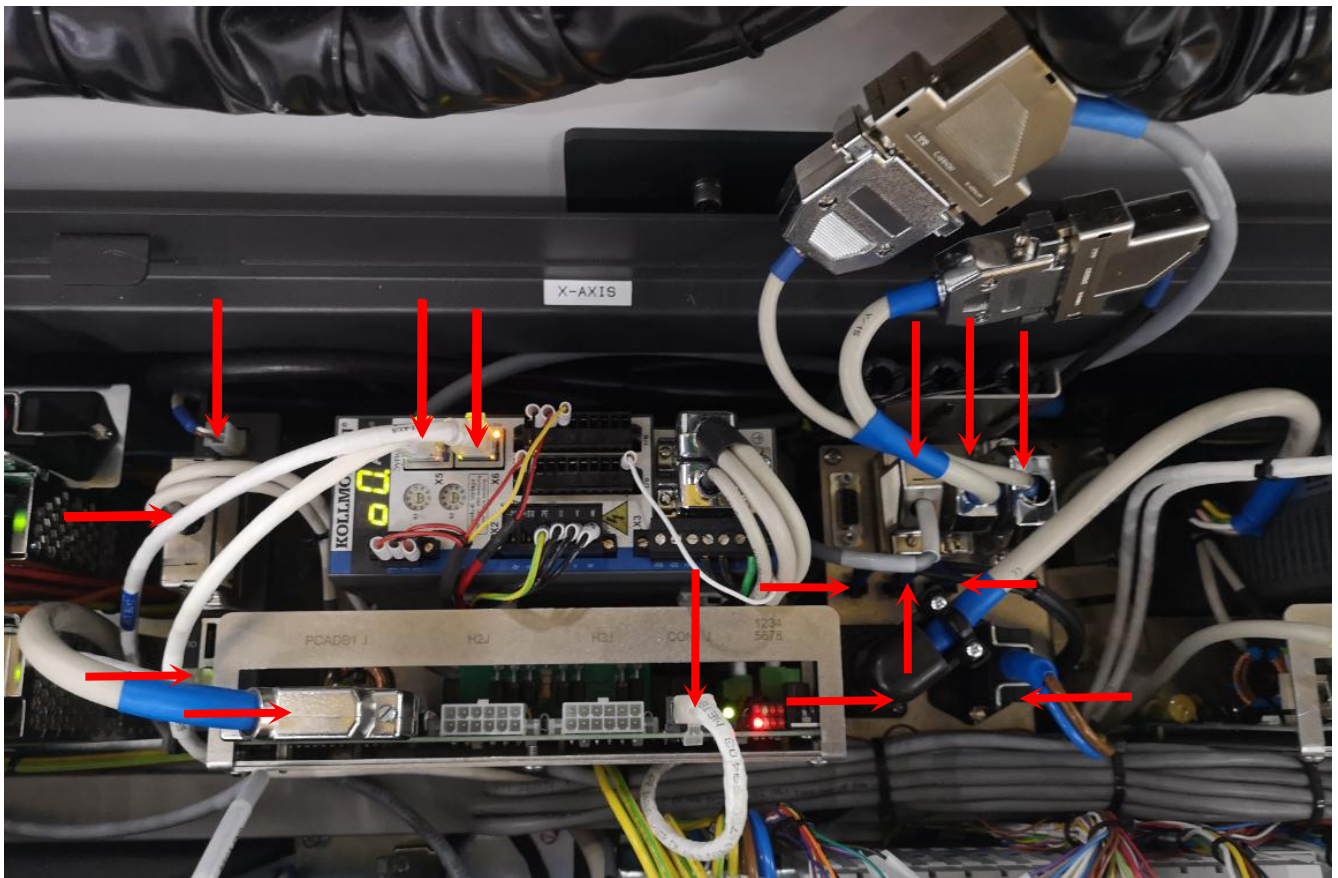


Figure 8 X-Axis AKD Kollmorgen

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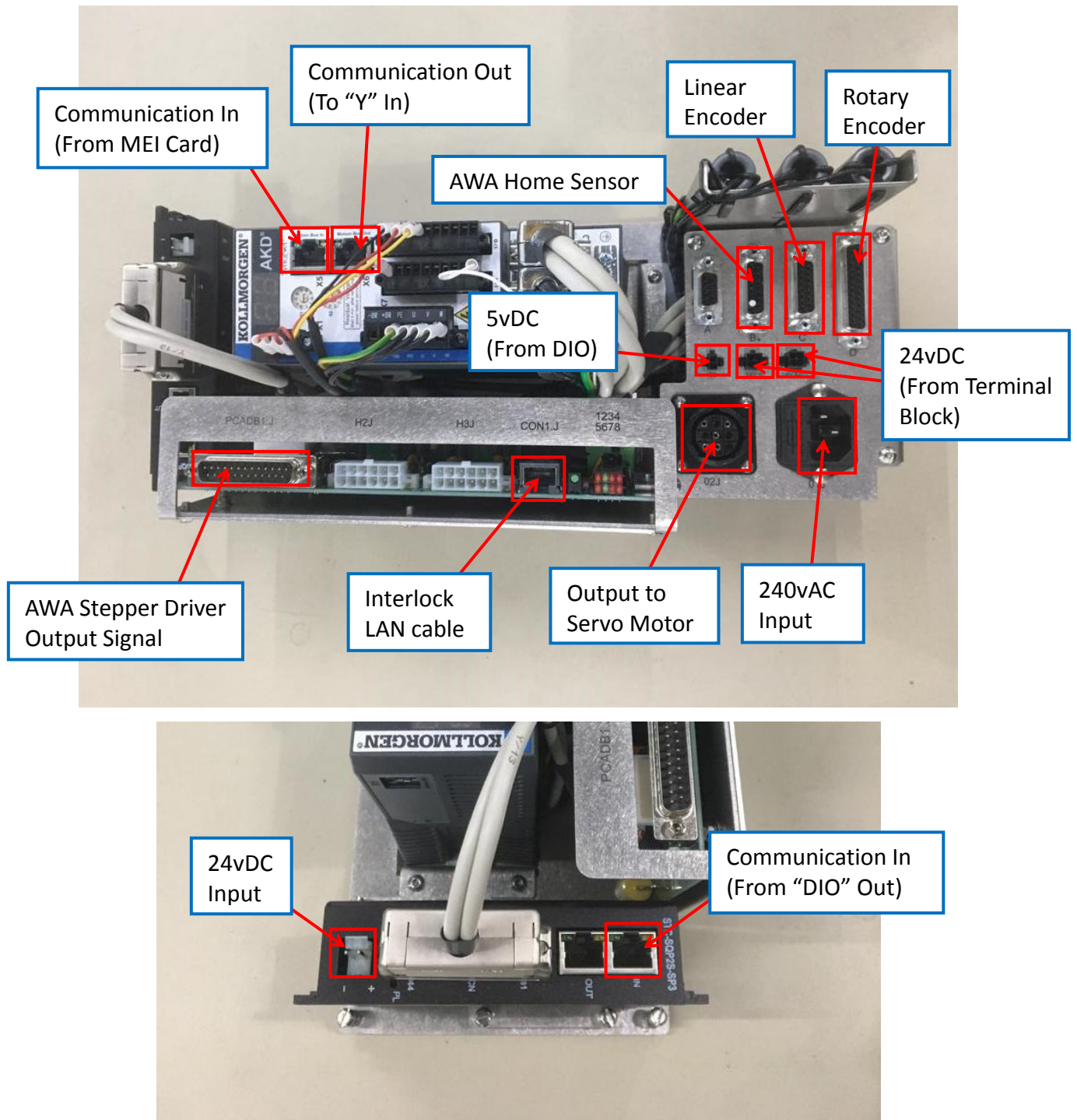


Figure 9 X-Axis AKD Kollmorgen

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X-Axis AKD Kollmorgen Drive Upgrade Procedure

1. Prepare a LAN (CAT5) cable with 1.5 meter length.



Figure 10 LAN (CAT5) cable

2. Connect LAN (CAT5) cable to Atom PC on board LAN port (origin empty port).

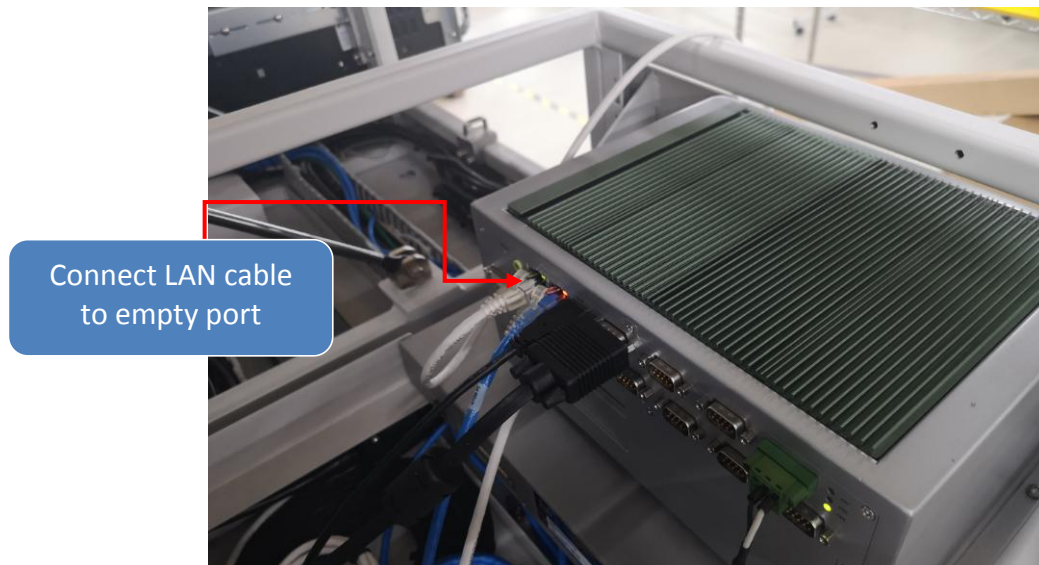


Figure 11 Connect LAN Cable to Atom PC on board LAN port

3. Connect other end of the LAN (CAT5) cable to the X-Axis AKD Kollmorgen.

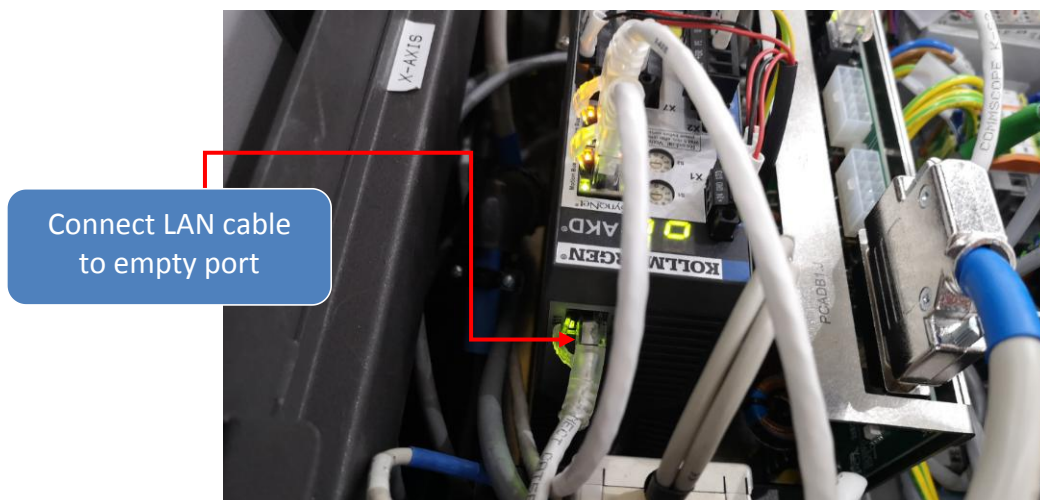


Figure 12 Connect LAN Cable to X-Axis AKD Kollmorgen

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4. Turn on SSCA power.
5. At main server, launch “Remote Desktop Connection” application.

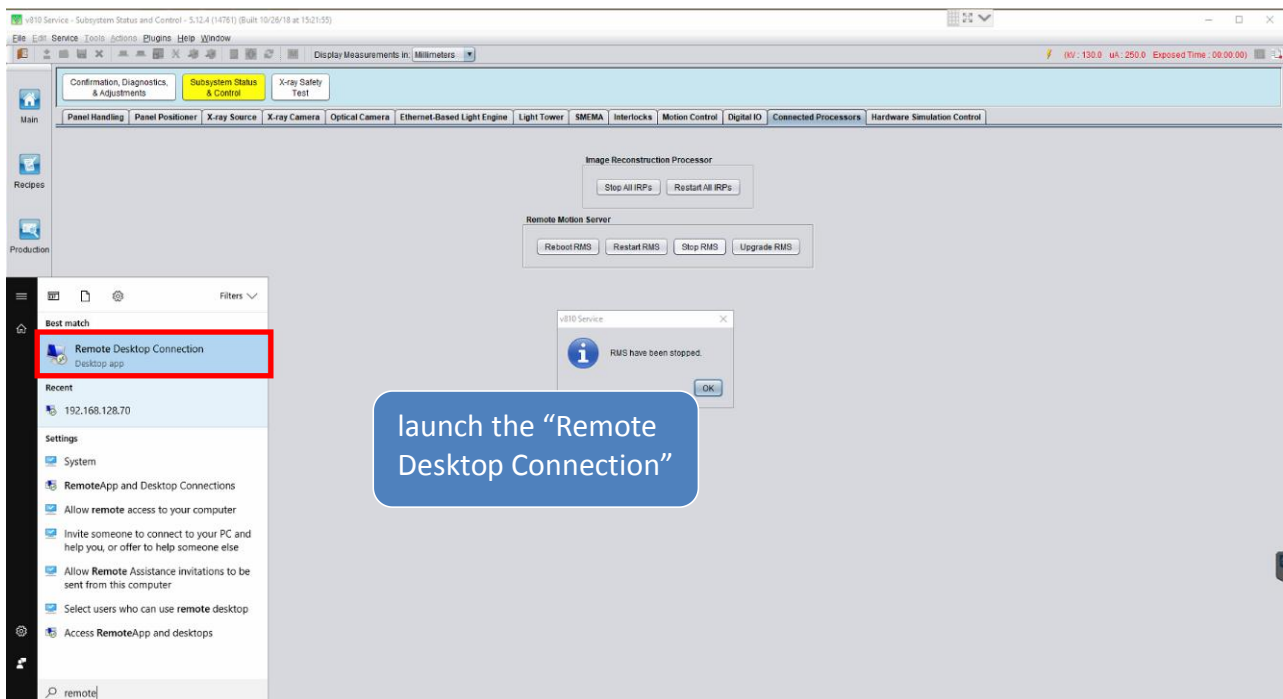


Figure 13 Launch Remote Desktop

6. Remote desktop to Atom PC with IP Address **192.168.128.70**.
Remark: Username, **Administrator**, Password, **Please!**
7. Launch “Kollmorgen WorkBench” application with clicking the Icon at Desktop. Click the “New Project”.
(Remark, the driver pre-installation would be require)

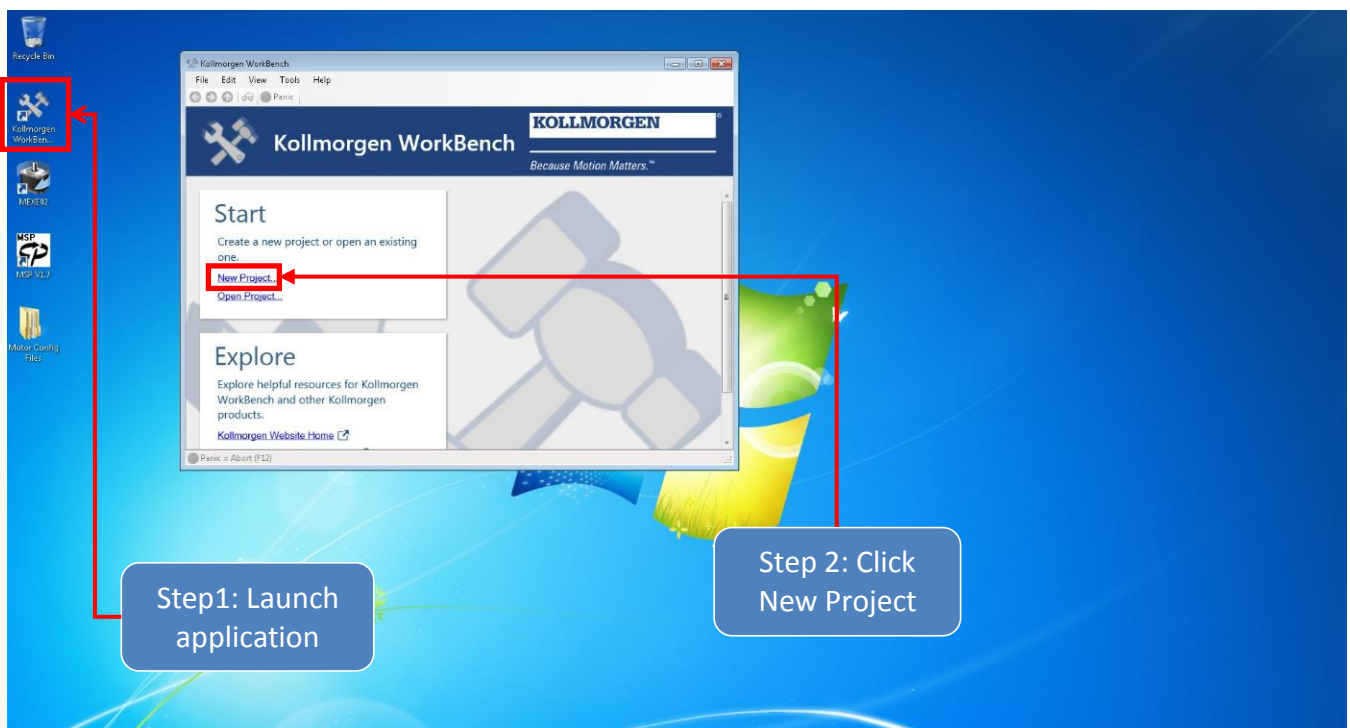


Figure 14 Launch Kollmorgen WorkBench

8. The interface below will shown, select the kollmorgen drive listed and click “Connect”.

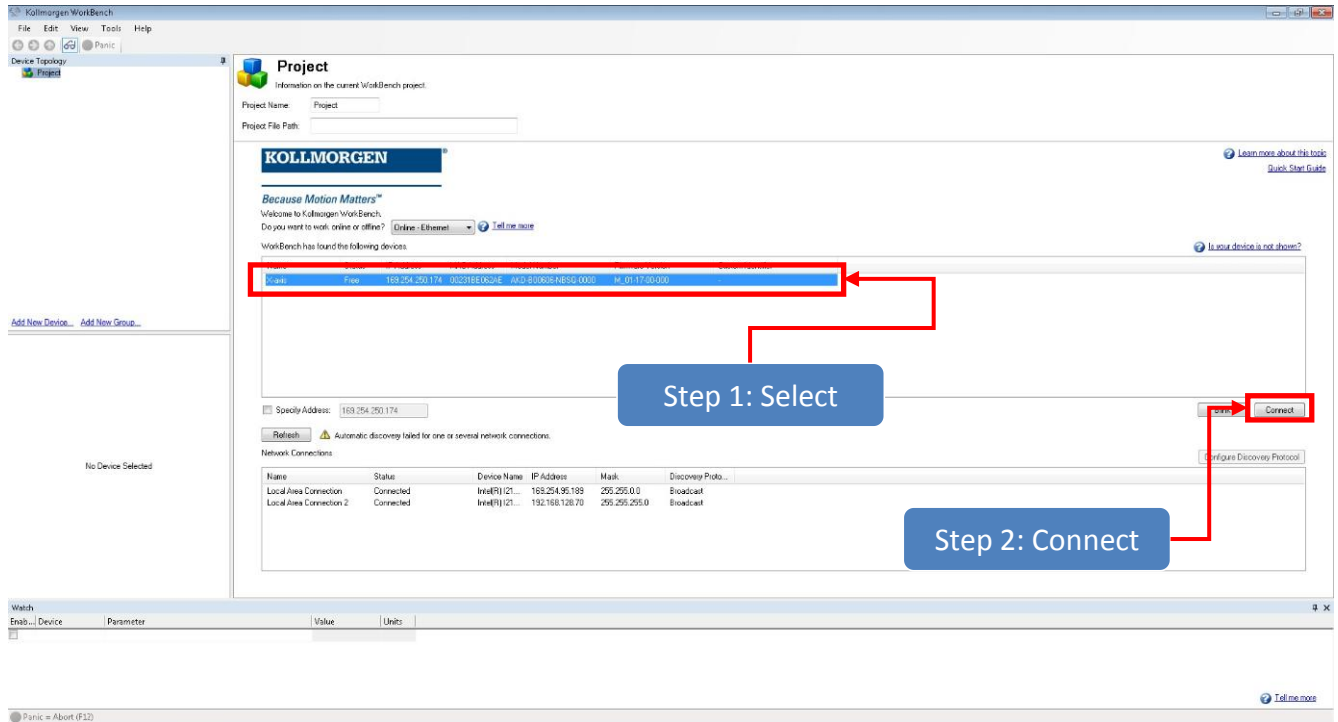


Figure 15 Connect to Kollmorgen Drive

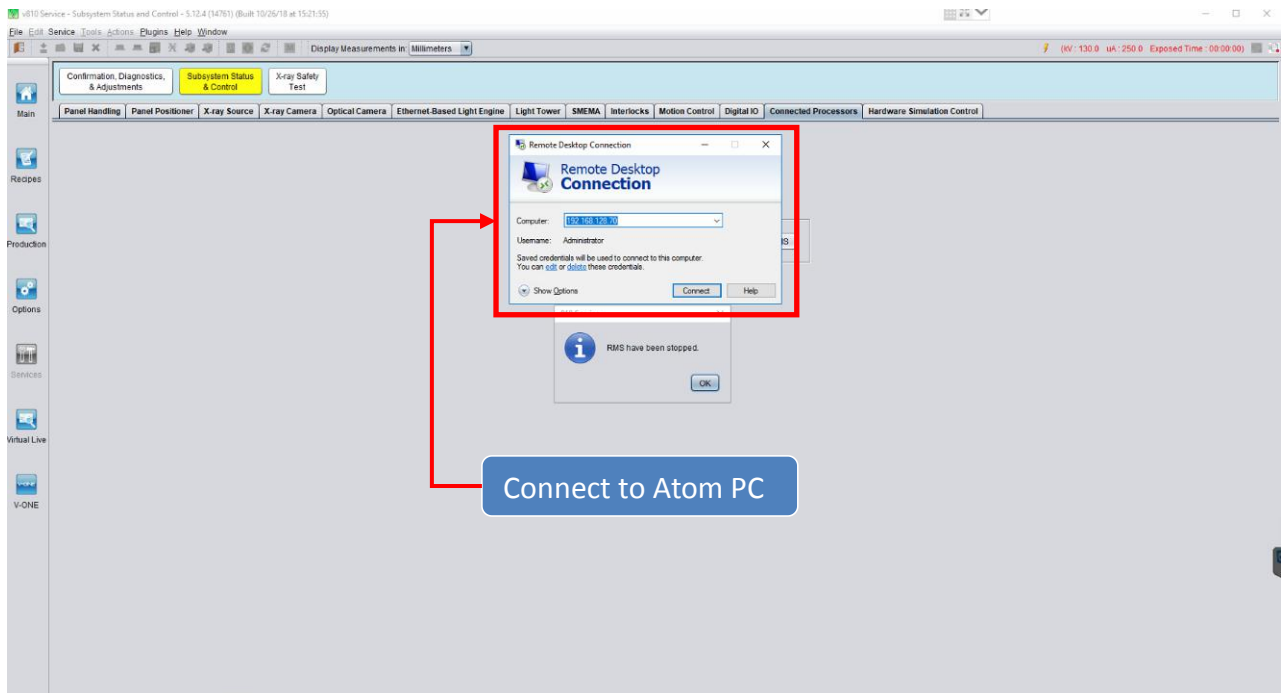


Figure 16 Launch Remote Desktop

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Upgrade X-Axis Kollmorgen Drive Firmware Update

1. The interface below will shown. Rename the drive with "X-Axis".
2. Click "More" continues with "Update".

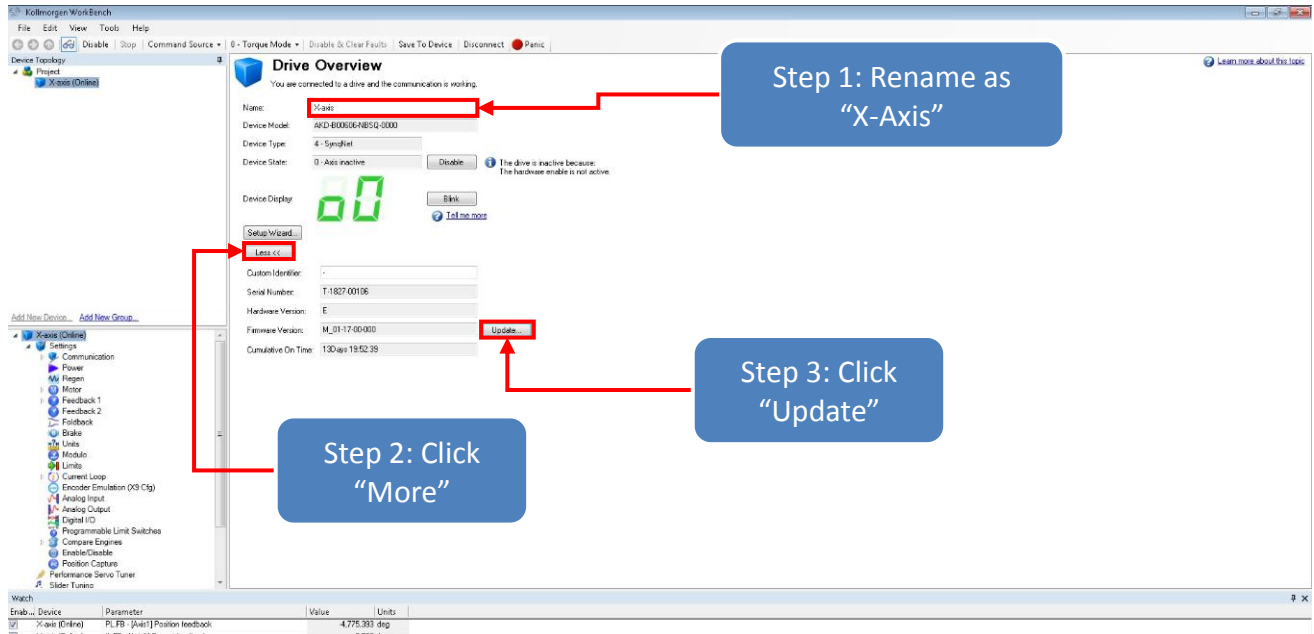


Figure 17 Log in to connected drive

3. Software will remind user that driver name has been changed and ask user whether to save this to device or not. Select "Yes" to continue.
4. Go to path "C:\Program Files\AXI System\v810\5.0\<Config>\motionControl\motionDrive\synqNet\AXI AKD Tuning Files".
5. Select "AKD-B-NBSQ-V01-17-00-000.i00" and click "Open".

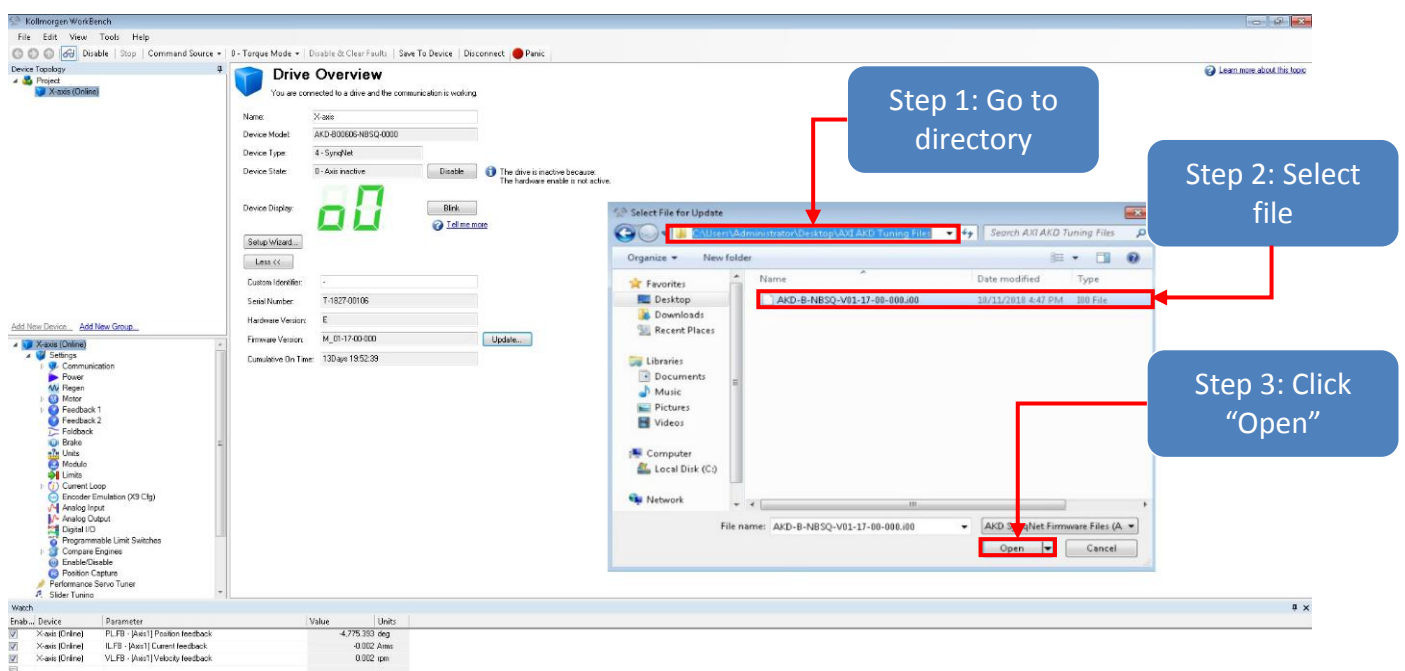


Figure 18 Select correct firmware file

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- The software will appear as loading status. It will takes few minutes to load, please wait until it finish and the driver status panel will appear again. This means the firmware is loaded successfully.

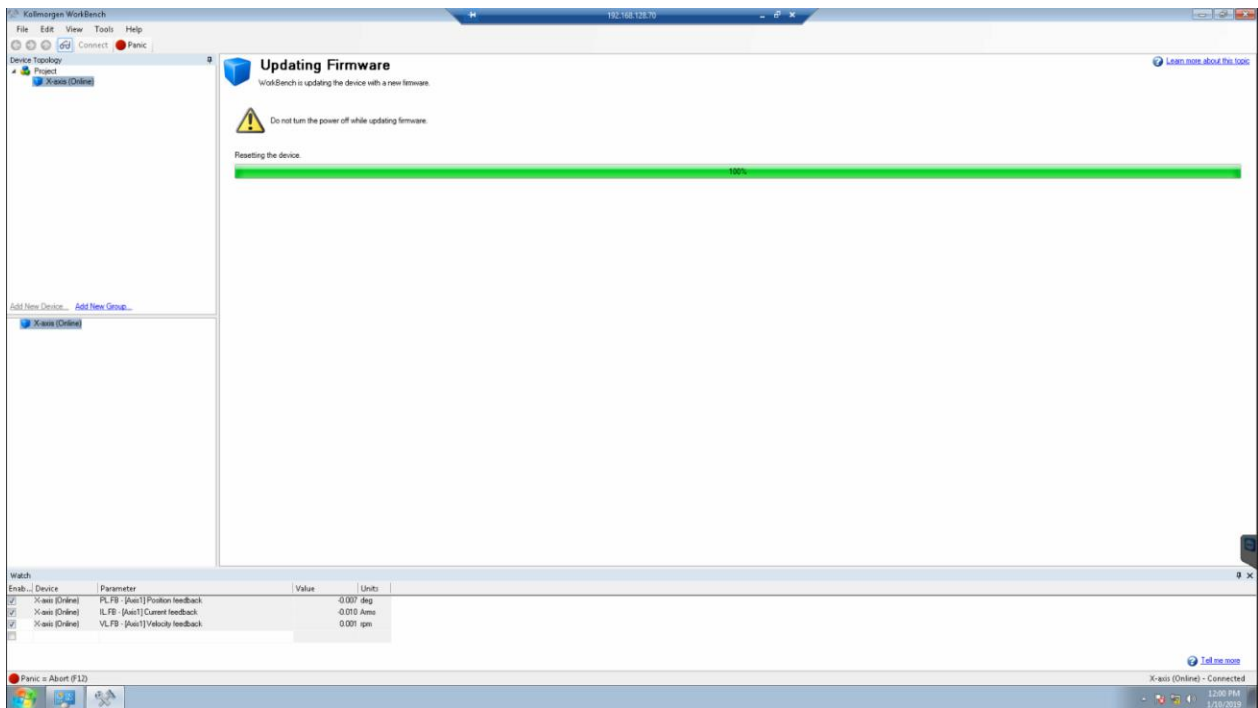


Figure 19 Complete firmware update

Upgrade X-Axis Kollmorgen Drive Parameter Update

- At the same Kollmorgen WorkBench” application, click “Tools” > “X Axis Online” > “Load Parameter”.

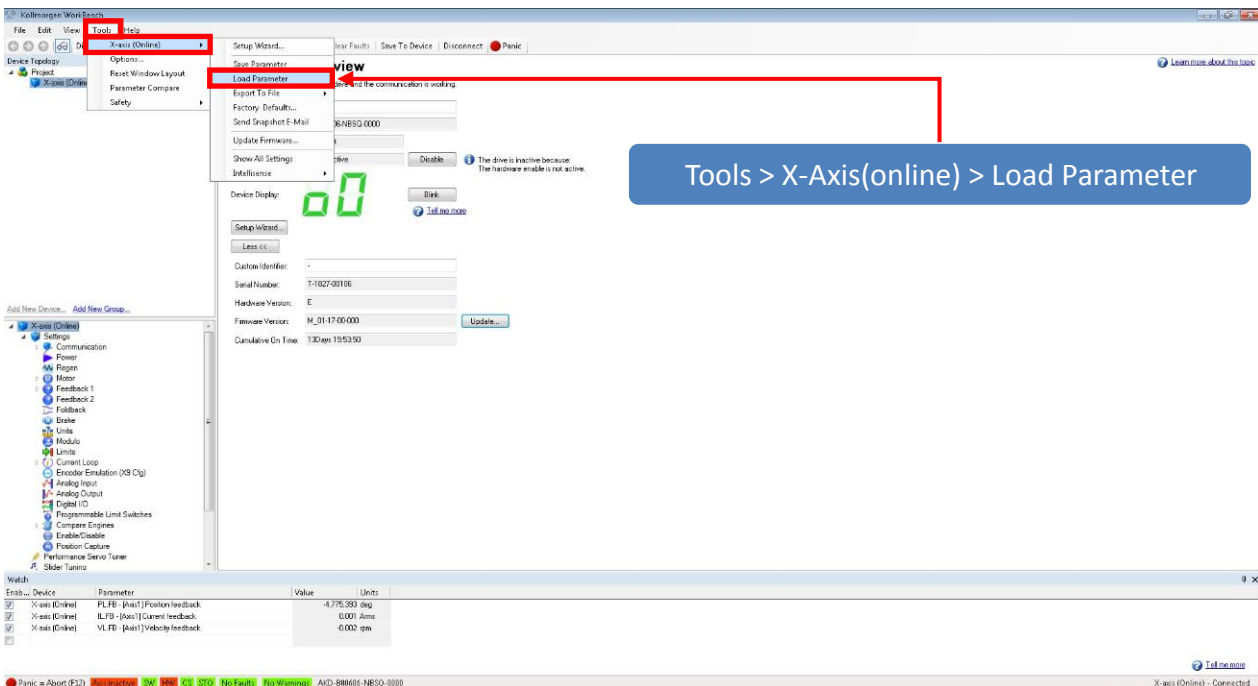
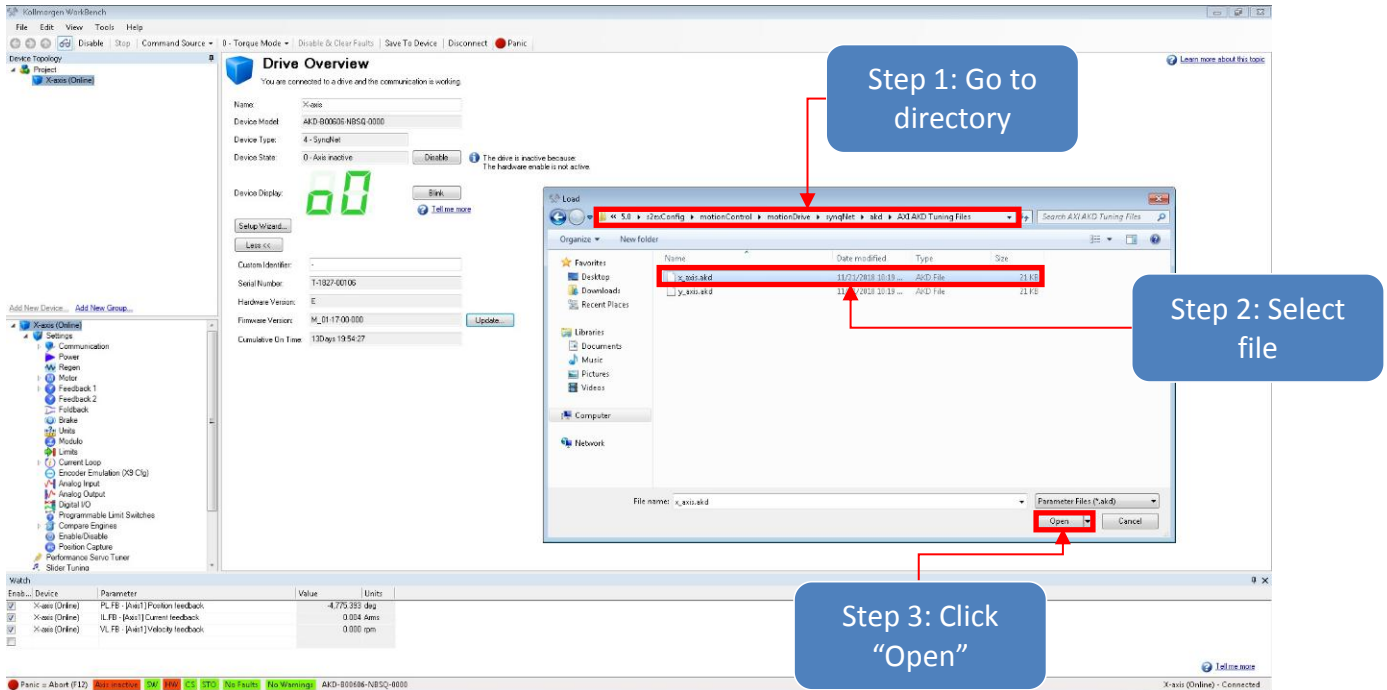


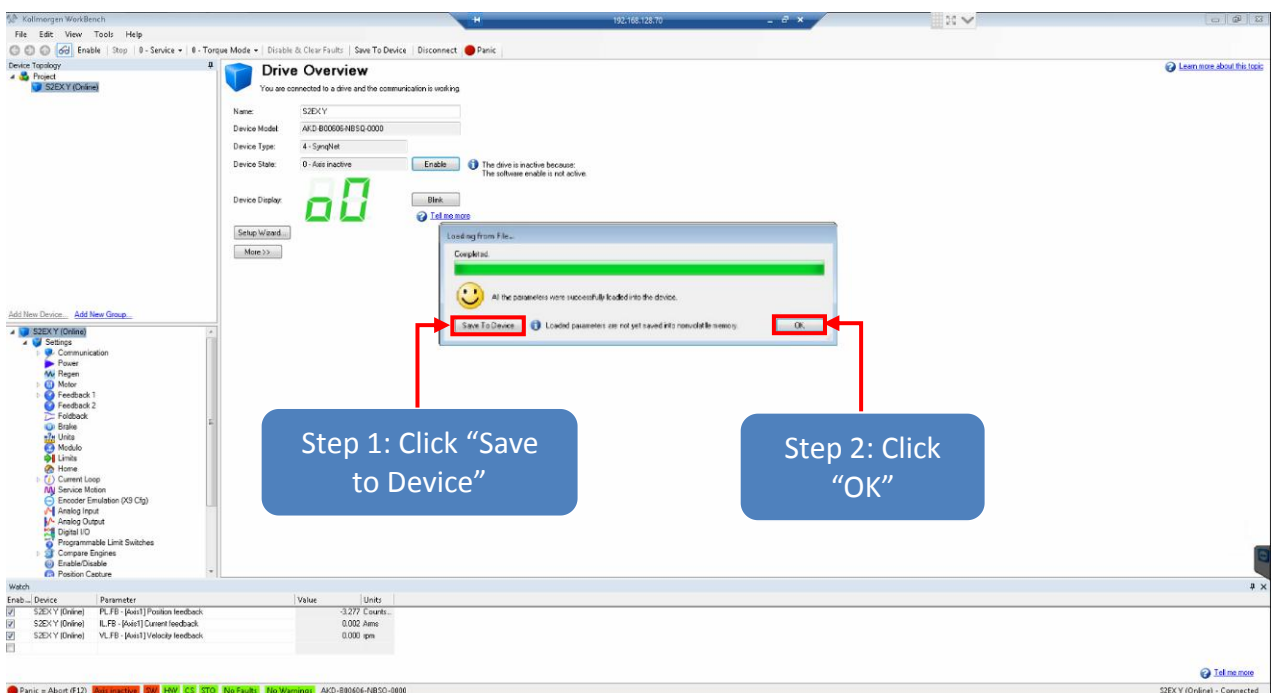
Figure 20 Load parameter

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- Go to path “C:\Program Files\AXI System\v810\5.0\<Config>\motionControl\motionDrive\synqNet\AXI AKD Tuning Files”.
- Select the “x_axis.akd” and click “Open”.



- A loading dialog will prompt out.
- When the parameter loaded successfully, a smiley face will show up.
- Click “Save To Device” button.
- Click “OK” to close the window.



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8. Once complete, exit “Kollmorgen WorkBench” and click “No”. **(Exit without save project)**
9. Close all the window.

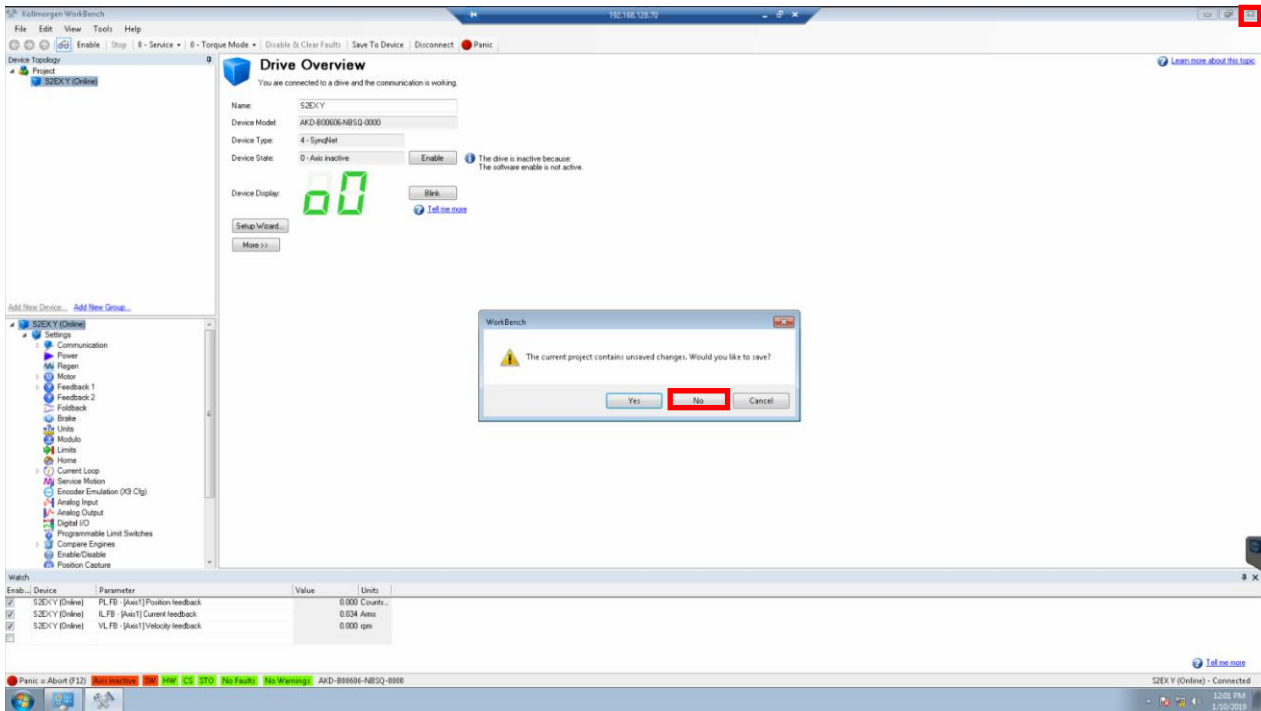


Figure 23 Save parameter to X-Axis AKD Driver

10. Disconnect & remove the external LAN cable from Atom PC and AKD driver.
11. Turn on SSCA & X-Ray source.
12. Open V810 GUI, go to Connected Processor, click Stop > Upgrade > Restart RMS.
13. Go to Motion Control tab, perform “Upgrade firmware”.
14. Click “Initialize” once complete upgrade.
15. Run 1 loop of CD&A.